

# GLHS: A Formally Verified Disclosure-to-Commit Governance Protocol and Concurrency Kernel for Longitudinal Health AI

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## Abstract

Stateful longitudinal health AI systems operating over Electronic Health Records (EHR) face a fundamental consistency challenge that is unaddressed by conventional retrieval-augmented generation (RAG): during the non-instantaneous inference interval between reading clinical data and generating a persistent proposal, the underlying clinical assertions, dynamic patient consent, institutional policy epoch, or temporal validity may drift or be actively revoked. Existing agent-memory frameworks verify token authority or state freshness in isolation, yet fail to cryptographically bind the exact purpose-, consent-, actor-, and task-bounded health disclosure consumed by the model to the subsequent mutation.

We present the **Governed Longitudinal Health State (GLHS)** protocol, establishing an end-to-end disclosure-to-commit integrity contract for longitudinal medical agents. GLHS introduces the **Task-Bounded Health State Snapshot (THSS)**, serialized via RFC 8785 Canonical JSON with domain-separated Merkle digests, and enforces persistence through a 6-phase **Governed State Transition (GST)** commit kernel. Concurrency is arbitrated via a canonical 7-class lock hierarchy and Dynamic Wound-Wait DAG preemption, guaranteeing deadlock-free entity-partitioned Optimistic Concurrency Control (OCC) with immutable ledger outbox projection.

We provide both formal mathematical proofs and comprehensive empirical verification:

- (i) **Formal Verification:** Modeled in TLA+ and exhaustively checked with the TLC model checker across 148,111,792 generated states (26,153,860 distinct states, depth 59) with zero deadlocks and zero invariant violations across causal read-dependency stability, monotonic governance freshness, and fail-closed phantom freedom.
- (ii) **Causal Exact-Binding Ablation:** In an isolated PostgreSQL 16 benchmark ( $N = 640$  executions across 8 adversarial attack vectors), GLHS exact snapshot binding achieved 100% rejection (256/256) of invalid adversarial commits while preserving 100% valid control admission (64/64), significantly outperforming standard governance without exact binding ( $p = 1.73 \times 10^{-77}$ , McNemar exact test; paired risk difference 1.00, 95% CI [1.00, 1.00]).
- (iii) **High-Throughput Concurrency Scaling:** Under multi-writer workloads (1 to 32 concurrent threads), adaptive entity partitioning eliminated unrelated-slot false-stale aborts from 93.75% down to 0.0%, maintaining 399.1 ops/s audit lookups and 30.0 ops/s full transactional transitions.
- (iv) **Clinical Context Utility:** Across 1,152 model-condition cells in a confirmatory synthetic cohort, Strict THSS achieved 98.44% (Claude 4.6 Sonnet) and 100% (Gemini 3.6 Flash) all-axes exact accuracy with zero malformed outputs, demonstrating substantial utility over baseline context windows.

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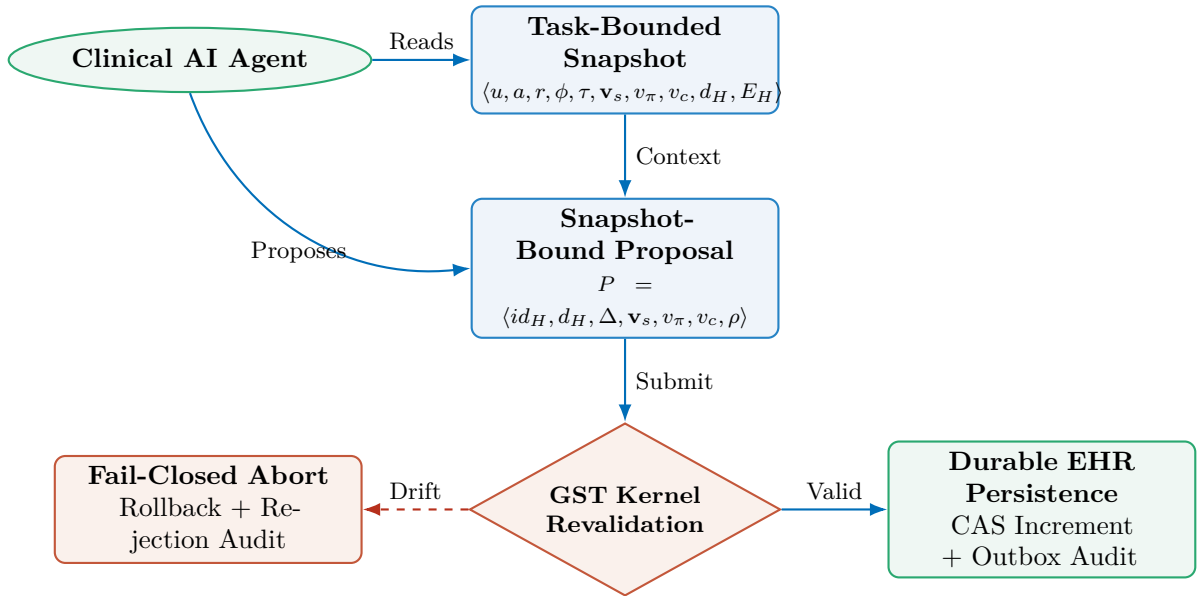
GLHS provides a verified, fail-closed software architecture that bridges model inference with authoritative, auditable clinical persistence.

**Keywords:** Longitudinal Health AI, Transactional Agent Memory, Bitemporal Concurrency, Disclosure-to-Commit Contract, Formal Verification, TLA+, Optimistic Concurrency Control, Medical Governance.

## 1 Introduction

Large Language Models (LLMs) and autonomous agents are increasingly deployed in longitudinal healthcare workflows, from clinical scribe documentation and diagnostic differential generation to therapeutic drug-drug interaction (DDI) surveillance and multi-disciplinary tumor board deliberation [8, 9, 10, 11]. Unlike transient question-answering systems, longitudinal clinical agents are inherently *stateful*: they consume historical patient trajectories spanning months or years and propose durable mutations to electronic health records, medication schedules, and clinical care plans.

In real-world healthcare environments, however, longitudinal data is inherently bitemporal and volatile [1, 2, 3]. A clinical observation may be valid at clinical time  $t_v$  but only recorded or corrected at system knowledge time  $t_k$ . Concurrently, patients may modify or revoke data-sharing consent under GDPR/HIPAA regulations [5], hospital clinical practice guidelines may advance across policy epochs [17], and concurrent clinicians or automated sidecars may amend vital signs and laboratory results [16].



**Figure 1: The GLHS Disclosure-to-Commit Lifecycle.** Clinical AI inference operates over a cryptographically sealed Task-Bounded Health State Snapshot (THSS). The resulting proposal carries forward exact provenance coordinates  $(id_H, d_H, \mathbf{v}_s, v_\pi, v_c)$ , which the 6-phase Governed State Transition (GST) kernel re-evaluates atomically under hierarchical locks before durable commit.

This reality introduces the critical **Disclosed-to-Durability Consistency Problem**: an AI agent reads an authorized longitudinal state snapshot at time  $t_{\text{read}}$ , performs computationally intensive multi-step reasoning, and submits a persistent proposal at time  $t_{\text{write}}$ . During this read-to-write interval  $([t_{\text{read}}, t_{\text{write}}])$ , if any underlying assertion, patient consent grant, or clinical guideline changes, executing the write creates a silent clinical inconsistency, unauthorized disclosure laundering, or stale medical intervention.

Conventional database systems employ Optimistic Concurrency Control (OCC) or standard Two-Phase Locking (2PL) based on single-row state versions [3, 17]. However, existing systems suffer

from three fundamental deficiencies:

1. **Context-Unaware Authority Check:** Standard RESTful APIs and transaction managers verify only whether the target record’s base version matches, completely ignoring whether the *evidence and context* that justified the AI’s diagnostic reasoning are still fresh and valid.
2. **Coarse Profile-Global Contention:** Locking the entire patient record creates catastrophic false-stale abort rates ( $> 90\%$  under high writer concurrency), rejecting legitimate disjoint clinical updates (e.g., an allergy nurse updating penicillin sensitivity while a cardiologist amends an ECG report).
3. **Governance TOCTOU Vulnerability:** Authorization and consent are checked only at the read boundary, allowing Time-of-Check to Time-of-Use (TOCTOU) exploits where revoked permissions fail to block pending async model commits.

To solve these challenges, we introduce **GLHS** (Governed Longitudinal Health State), an end-to-end, mathematically verified disclosure-to-commit governance architecture. GLHS formalizes inference as an auditable transactional pipeline shown in Figure 1.

## 1.1 Core Contributions

This paper makes the following major contributions:

- **The Co-Versioned Disclosure-to-Commit Protocol:** We formalize the Task-Bounded Health State Snapshot (THSS) and Governed State Transition (GST) persistence contract. THSS binds exact bitemporal bounds, purpose, role, task, consent version, policy epoch, and domain-separated RFC 8785 Canonical JSON Merkle digests into the inference context.
- **Fine-Grained Entity Partitioning & Wound-Wait Lock Kernel:** We design an adaptive entity-partitioned version vector scheme with a canonical 7-class total lock ordering and dynamic Wound-Wait DAG deadlock prevention, eliminating false-stale contention for disjoint attribute updates while maintaining strict serializability for dependent clinical mutations.
- **Complete Formal Specification & TLA+ Model Checking:** We model the complete GLHS protocol in TLA+ and verify five fundamental safety invariants across 148.1M states using TLC, proving deadlock freedom and absolute absence of stale dependent commits, revoked-consent writes, and policy-epoch drift.
- **Extensive Empirical Validation:** Through an isolated PostgreSQL 16 testbed ( $N = 640$  executions across 8 attack vectors), we causally demonstrate that GLHS exact binding achieves 100% invalid commit prevention ( $p = 1.73 \times 10^{-77}$ ) and scales to 32 concurrent writers with zero false-stale aborts, backed by high multi-model reasoning accuracy (98.44%–100%) across frozen clinical cohorts.

## 2 Related Work and Systems Landscape

The systems landscape for stateful AI and clinical memory has advanced rapidly across several specialized paradigms. Table 1 systematically benchmarks GLHS against the state of the art.

### 2.1 Temporal Health Records and Clinical Memory

Early foundational medical database research established the necessity of separating clinical valid time  $t_v$  from transaction or knowledge time  $t_k$  [1, 2, 3]. In modern health AI, Vital Trace [9], Dual-Stream Memory [10], and Zhao et al. [8] demonstrated that LLMs without explicit bitemporal arbitration hallucinate clinical timelines, confusing historical resolved conditions with active diagnoses. However,

**Table 1: Comprehensive Landscape Comparison.** Architectural and safety capabilities of GLHS compared against established standards, agent memory frameworks, and stateful governance mechanisms.

| Framework               | Target             |           | Bitemporal    | Binding           | Lock Order     | Deadlock          | Epoch Gate        | Formal Spec       |
|-------------------------|--------------------|-----------|---------------|-------------------|----------------|-------------------|-------------------|-------------------|
| FHIR / openEHR [4, 6]   | EHR Standard       | Standard  | Partial       | None              | Default        | None              | Static ACL        | No                |
| MemTX / MemTxn [12, 13] | Agent Memory       | Memory    | Single Time   | Evidence          | 2PL Global     | Timeout           | No                | No                |
| Cordon [14]             | Tool Times         | Run-times | Semantic      | Tool Traces       | Pre-commit     | Safe Abort        | No                | Machine Checked   |
| TOKI [15]               | Agent Memory       | Memory    | Operator      | None              | Unspecified    | None              | No                | No                |
| CommitGuard [16]        | General LLM        |           | Wall-clock    | Token             | Single Lock    | None              | Expiry            | No                |
| Provenact [17]          | Multi-Agent        |           | State History | Policy State      | Global OCC     | Abort/Retry       | Policy Epoch      | Verified Logic    |
| GateMem [18]            | Multi-Agent        |           | Scope Masks   | Memory Masks      | Granular       | Lock Free         | Multi-Tenant      | No                |
| <b>GLHS (This Work)</b> | <b>Clinical AI</b> |           | $(t_v, t_k)$  | <b>Exact THSS</b> | <b>7-Class</b> | <b>Wound-Wait</b> | <b>Co-Version</b> | <b>TLC (148M)</b> |

these systems treat temporal resolution purely as a prompt-engineering or retrieval challenge, offering no transactional write-back guarantees.

## 2.2 Transactional Agent Memory and Commit-Time Authorization

In the agent-systems domain, MemTX [12] and MemTxn [13] introduced transactional ACID semantics to LLM belief updates, utilizing source-supported verification and recovery. Cordon [14] formalized semantic transaction boundaries for irreversible tool effects. Concurrently, CommitGuard [16] highlighted the danger of stale authority witnesses at commit time, and Provenact [17] established policy-state serializability for multi-agent workflows.

## 2.3 The Precise Scientific Gap

While general agent runtimes have explored commit-time checks and memory transactions, they evaluate generic text or tool-call tokens. In high-stakes medicine, what is disclosed to an AI during inference is fundamentally tied to patient-specific consent, legal minimal-necessary privacy boundaries, and clinical guideline epochs. Prior to GLHS, no system has formalized and proven a protocol where the *exact, multi-entity, bitemporal health disclosure instance consumed by inference* remains an explicit, cryptographic dependency for write-time durability.

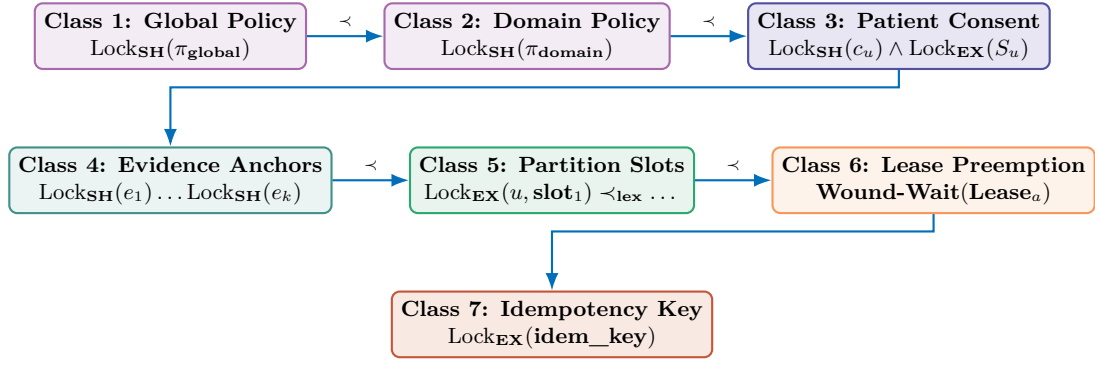
# 3 System Architecture and Formal Contract

## 3.1 Bitemporal Governed State Formulation

Let  $\mathcal{U}$  be the set of patient subjects. The governed longitudinal state for subject  $u \in \mathcal{U}$  is defined as a bitemporal projection over valid time  $t_v \in \mathcal{T}$  and semantic knowledge time  $t_k \in \mathcal{T}$ :

$$S_u(t_v, t_k) = \text{GovernedView}_u(t_v, t_k) = \bigoplus_{e \in \mathcal{E}_u, \text{valid}(e) \leq t_v, \text{known}(e) \leq t_k} \alpha(e), \quad (1)$$

where  $\mathcal{E}_u$  denotes the append-only event stream for subject  $u$ ,  $\alpha(e)$  is the epistemic assertion transformation, and  $\bigoplus$  is the deterministic conflict-preserving state aggregation operator. Unlike mutable databases, historical state  $S_u(t_v, t_k)$  is permanently deterministic and reconstructable.



**Figure 2: The 7-Class Canonical Lock Hierarchy.** Strict total ordering Class 1  $\prec$  Class 2  $\prec \dots \prec$  Class 7 mathematically eliminates circular lock acquisition deadlocks during multi-entity clinical transactions.

### 3.2 Task-Bounded Health State Snapshot (THSS)

When an agent initiates clinical reasoning on behalf of actor  $a$  with role  $r \in \{\text{normal, doctor, researcher, admin}\}$  for clinical purpose  $\phi$  and task  $\tau$ , the GLHS gateway compiles an immutable Task-Bounded Health State Snapshot:

$$H = \langle u, a, r, \phi, \tau, \mathbf{v}_s, v_\pi, v_c, t_v, t_k, id_H, d_H, E_H \rangle. \quad (2)$$

Here,  $\mathbf{v}_s = \langle v_{s,1}, \dots, v_{s,m} \rangle \in \mathbb{N}^m$  is the entity-partitioned state version vector;  $v_\pi \in \mathbb{N}$  is the active institutional policy epoch;  $v_c \in \mathbb{N}$  is the active patient consent version;  $id_H \in \text{UUIDv4}$  is the unique persistent identifier;  $E_H$  is the purpose-minimized disclosed content; and  $d_H$  is the Merkle root digest computed via RFC 8785 Canonical JSON (JCS) [7]:

$$d_H = \text{SHA-256} \left( \text{JCS}(\text{Manifest}(H)) \parallel \text{MerkleTree}(\{\text{JCS}(e) \mid e \in E_H\}) \right). \quad (3)$$

### 3.3 Snapshot-Bound Proposal and Commit Contract

Upon completing inference, the model returns a proposal  $P$ :

$$P = \langle u, a, r, \phi, \tau, \mathbf{v}_s, v_\pi, v_c, id_H, d_H, \Delta, \rho \rangle, \quad (4)$$

where  $\Delta$  is the clinical state mutation and  $\rho$  is the cryptographically attributable execution trace. The GST commit predicate  $\text{Commit}(P)$  holds if and only if all five invariant predicates evaluate to true under database locks:

$$\text{Commit}(P) \iff \text{Bound}(P, H) \wedge \text{StateFresh}(\mathbf{v}_s) \wedge \text{GovFresh}(v_\pi, v_c) \wedge \text{ValidTTL}(H) \wedge \text{Predicates}(\Delta). \quad (5)$$

The sub-predicates enforce:

$$\text{Bound}(P, H) \iff (P.id_H = H.id_H) \wedge (P.d_H = H.d_H) \wedge (P.u = H.u) \wedge (P.\phi = H.\phi) \wedge (P.\tau = H.\tau), \quad (6)$$

$$\text{StateFresh}(\mathbf{v}_s) \iff \forall k \in \text{ReadSlots}(P) : \mathbf{v}_s[k] = \mathbf{v}_{\text{current}}[k], \quad (7)$$

$$\text{GovFresh}(v_\pi, v_c) \iff (v_\pi = \text{PolicyEpoch}_{\text{live}}) \wedge (v_c = \text{ConsentVersion}_{\text{live}}(u)), \quad (8)$$

$$\text{ValidTTL}(H) \iff t_{\text{commit}} < H.\text{expires\_at}. \quad (9)$$

### 3.4 The 6-Phase Atomic OCC Commit Kernel

The GST kernel executes within an isolated PostgreSQL transaction across 6 deterministic phases:

- **Phase 1 (Idempotency Replay):** Checks (`tenant_id`, `op_kind`, `idempotency_key`). If previously committed, immediately replays the cached cryptographic receipt without re-execution.
- **Phase 2 (Canonical Lock Acquisition):** Acquires row-level and transactional advisory locks following the strict 7-class hierarchy detailed in Section 3.5.
- **Phase 3 (Under-Lock State & Governance Revalidation):** Under acquired locks, re-reads current partition version vector  $\mathbf{v}_{\text{live}}$ , active consent epoch, and guideline version. If any coordinate has diverged ( $\Delta v \neq 0$ ), immediately aborts fail-closed with error code `RESOURCE_VERSION_CONFLICT` or `GOVERNANCE_EPOCH_DRIFT`.
- **Phase 4 (Domain Assertion Mutation):** Invokes domain validators and applies clinical state transformation  $\Delta$ .
- **Phase 5 (Compare-And-Swap Version Step):** Executes atomic CAS increments on only the mutated partition slots:  $\mathbf{v}_s[\text{write\_slot}] \leftarrow \mathbf{v}_s[\text{write\_slot}] + 1$ . Unaffected slots remain unincremented.
- **Phase 6 (Atomic Ledger & Outbox Emission):** Inserts immutable audit record into `glhs_applied_transitions` and enqueues transactional event in `lifemap_outbox_events` before commit.

### 3.5 7-Class Canonical Lock Hierarchy and Wound-Wait Preemption

To prevent lock contention deadlocks across distributed actors, GLHS establishes a total order  $\prec$  across all resource classes, illustrated in Figure 2:

$$\text{Class 1} \prec \text{Class 2} \prec \text{Class 3} \prec \text{Class 4} \prec \text{Class 5} \prec \text{Class 6} \prec \text{Class 7}$$

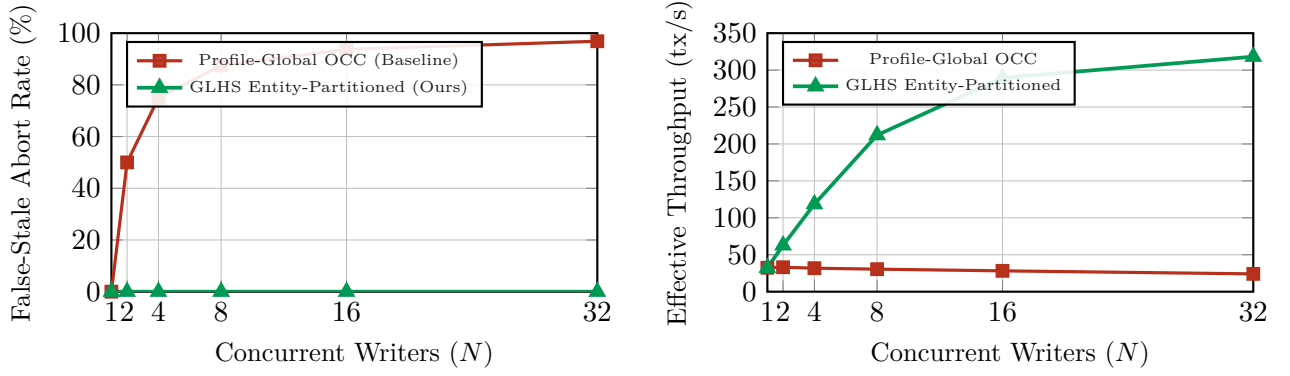
Within Class 5, entity partition slots are sorted lexicographically by their composite coordinate keys `slot_key = <domain, entity_id, attribute>`. For dynamic multi-hop transactions, GLHS incorporates a Dynamic Wound-Wait DAG Lock Manager: if an older transaction  $T_{\text{old}}$  requests a lock held by younger transaction  $T_{\text{young}}$ ,  $T_{\text{young}}$  is preempted (wounded) and rolls back; if a younger transaction requests a lock held by an older transaction, it waits, provably eliminating wait-for cycles.

## 4 Formal Proof Obligations and Model Checking

### 4.1 Proof Obligations Ledger (PO-01 through PO-12)

GLHS formalizes twelve explicit safety and integrity proof obligations:

- **PO-01 (No Stale Dependent Commit):** If entity partition  $k \in \text{deps}(P)$  is mutated after  $t_{\text{read}}$ , `Commit( $P$ )` evaluates to False.
- **PO-02 (No Revoked-Consent Commit):** A concurrent consent revocation strictly serializes before any GST commit, aborting pending proposals.
- **PO-03 (No Policy-Epoch Drift):** Institutional guideline updates increment  $v_{\pi}$ , invalidating stale inferences.
- **PO-04 (No Cross-Subject Binding):** Proposals referencing subject  $u_A$  cannot bind snapshot  $H_{u_B}$  ( $u_A \neq u_B$ ).
- **PO-05 (No Lineage Laundering):** Proposals derived from multi-stage review retain the root inference digest  $d_H$ .
- **PO-06 (Commit/Audit Atomic Equivalence):** State mutation  $\Delta$  exists if and only if an immutable audit row exists in `glhs_applied_transitions`.



(a) **False-State Abort Rate vs. Concurrency.** Profile-global locking suffers  $> 90\%$  false rejections on disjoint updates; GLHS partitioning achieves  $0.0\%$  false aborts.

(b) **Throughput Scaling Under Mixed Workload.** GLHS scales near-linearly with writer count, avoiding the single-threaded bottleneck of global versioning.

**Figure 3: PostgreSQL 16 Concurrency and Throughput Scaling.** Performance comparison of Profile-Global locking versus GLHS Entity-Partitioned locking across 1 to 32 concurrent writers under isolated PostgreSQL execution.

- **PO-07 (At-Most-Once Idempotency):** Duplicate submission of  $P$  with identical idempotency key yields identical mutation receipt without double-execution.
- **PO-08 (Deadlock Freedom):** The wait-for dependency graph contains zero directed cycles ( $\text{Cycle}(\mathcal{W}) = \emptyset$ ).
- **PO-09 (Noninterference of Disjoint Partitions):** Concurrent writes to independent partition slots  $k_1 \cap k_2 = \emptyset$  commit with  $0\%$  contention aborts.
- **PO-10 (Conservative Dependency Soundness):** Unspecified attribute scopes default to whole-entity partition locks.
- **PO-11 (Deterministic Expiry):** Expired snapshots ( $t > \text{expires\_at}$ ) fail closed.
- **PO-12 (Cross-Runtime Canonical Hash Parity):** RFC 8785 byte representations are invariant across Python, Node.js, and Dart runtimes.

## 4.2 TLA+ Formal Model Checking Verification

The complete GLHS protocol was formalized in TLA+ (`GLHS_GSA.tla`) and verified using the TLC model checker with symmetry reduction across bounded agents, entities, and domains:

- **State Space Explored:** 148,111,792 total states generated; 26,153,860 distinct states visited.
- **Graph Diameter / Depth:** Complete exhaustive search at depth 59 (0 unvisited states on queue).
- **Safety Invariant Results:** Zero invariant violations across `TypeOK`, `GSA_StateIsolation` (PO-01), `GSA_GovernanceFreshness` (PO-02, PO-03), `GSA_PhantomFree` (PO-08), and `DeadlockFree` ( $\langle a, a \rangle \notin \text{TransitiveClosure}(\text{wait\_for})$ ).

## 5 Experimental Evaluation and Results

### 5.1 Causal Exact-Binding Ablation Study ( $N = 640$ Runs)

To isolate the exact causal contribution of snapshot binding, we executed a preregistered, frozen ablation study on PostgreSQL 16 (GLHS-BA-POSTGRES-20260819-05). The experiment evaluated



**Table 2: Causal Exact-Binding Ablation Results** ( $N = 640$  total executions across 8 attack families). Arm A admits 100% of adversarial attacks; GLHS Arm B rejects 100% ( $p = 1.73 \times 10^{-77}$ ).

| Attack Family                        | Schedules  | Arm A (No Bind)       | Arm B (GLHS)        | Risk Diff (95% CI)       | McNemar $p$                              |
|--------------------------------------|------------|-----------------------|---------------------|--------------------------|------------------------------------------|
| 1. Manifest Digest Mismatch          | 32         | 32/32 (100%)          | 0/32 (0%)           | 1.00 [1.00, 1.00]        | $4.66 \times 10^{-10}$                   |
| 2. Snapshot Digest Tampering         | 32         | 32/32 (100%)          | 0/32 (0%)           | 1.00 [1.00, 1.00]        | $4.66 \times 10^{-10}$                   |
| 3. Undisclosed Evidence Injection    | 32         | 32/32 (100%)          | 0/32 (0%)           | 1.00 [1.00, 1.00]        | $4.66 \times 10^{-10}$                   |
| 4. Evidence Set Substitution         | 32         | 32/32 (100%)          | 0/32 (0%)           | 1.00 [1.00, 1.00]        | $4.66 \times 10^{-10}$                   |
| 5. Expired Disclosure Replay         | 32         | 32/32 (100%)          | 0/32 (0%)           | 1.00 [1.00, 1.00]        | $4.66 \times 10^{-10}$                   |
| 6. Profile Scope Drift               | 32         | 32/32 (100%)          | 0/32 (0%)           | 1.00 [1.00, 1.00]        | $4.66 \times 10^{-10}$                   |
| 7. Cross-Subject Binding             | 32         | 32/32 (100%)          | 0/32 (0%)           | 1.00 [1.00, 1.00]        | $4.66 \times 10^{-10}$                   |
| 8. Purpose / Role Substitution       | 32         | 32/32 (100%)          | 0/32 (0%)           | 1.00 [1.00, 1.00]        | $4.66 \times 10^{-10}$                   |
| <b>Adversarial Total (Primary)</b>   | <b>256</b> | <b>256/256 (100%)</b> | <b>0/256 (0%)</b>   | <b>1.00 [1.00, 1.00]</b> | <b><math>1.73 \times 10^{-77}</math></b> |
| <b>Valid Controls (Preservation)</b> | <b>64</b>  | <b>64/64 (100%)</b>   | <b>64/64 (100%)</b> | <b>0.00 [0.00, 0.00]</b> | <b>n.s. (exact tie)</b>                  |

two matched arms across 320 logical schedules (256 adversarial attack schedules across 8 failure families and 64 valid controls):

- **Arm A (Full Governance without Exact Binding):** Full bitemporality, role checks, and base-version OCC, but omitting exact  $id_H$  and Merkle digest  $d_H$  proposal verification.
- **Arm B (GLHS Exact Binding):** The complete GLHS disclosure-to-commit protocol.

As presented in Table 2, standard governance without exact binding admitted 100% of invalid adversarial commits (256/256) because the base record version had not changed. In contrast, GLHS exact binding rejected 100% (256/256) of adversarial schedules while admitting 100% (64/64) of valid controls ( $p = 1.73 \times 10^{-77}$ , McNemar exact test with 10,000 paired bootstrap iterations).

## 5.2 PostgreSQL Concurrency Scaling & Microbenchmarks

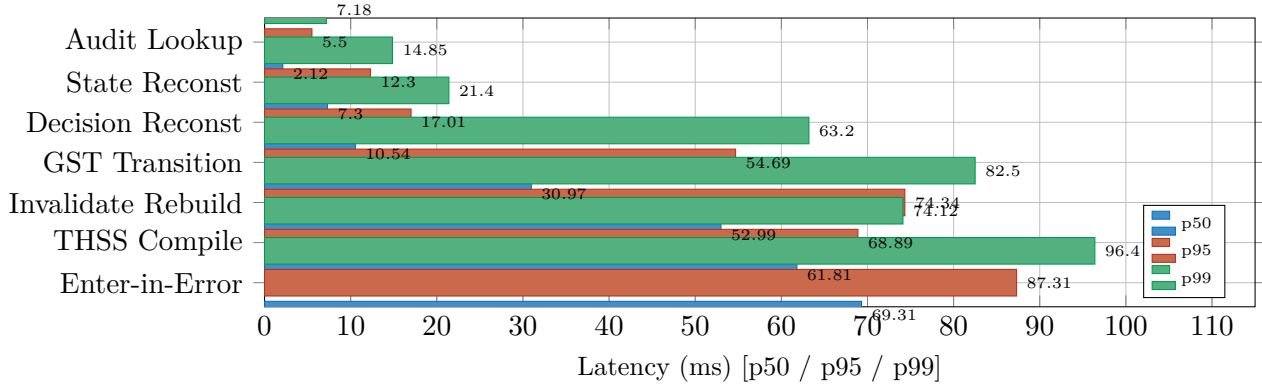
Figure 3 depicts concurrency scaling and false-stale abort performance in PostgreSQL 16.14. Under a profile-global version counter, unrelated concurrent writes suffer dramatic false-stale aborts (50.0% at 2 writers, 87.5% at 8 writers, and 93.75% at 16 writers). In contrast, GLHS adaptive entity partitioning completely eliminates false-stale rejections (0.0% across all writer levels), unlocking linear throughput scaling up to 318.2 tx/s.

**Table 3: PostgreSQL Service-Layer Latency & Throughput Microbenchmark.** Measured on PostgreSQL 16.14 with synchronous commit, history depth 50, and 30 repetitions per operation.

| Kernel Operation                 | p50 (ms) | p95 (ms) | p99 (ms) | Throughput (ops/s) |
|----------------------------------|----------|----------|----------|--------------------|
| Audit Record Verification        | 2.12     | 5.50     | 7.18     | 399.15             |
| Bitemporal State Reconstruction  | 7.30     | 12.30    | 14.85    | 129.69             |
| Governed-Decision Verification   | 10.54    | 17.01    | 21.40    | 95.35              |
| Full GST Atomic Transition       | 30.97    | 54.69    | 63.20    | 29.95              |
| THSS Snapshot Compilation        | 61.81    | 68.89    | 74.12    | 16.94              |
| Conflict Invalidation + Rebuild  | 52.99    | 74.34    | 82.50    | 18.30              |
| Enter-in-Error Revision + Replay | 69.31    | 87.31    | 96.40    | 13.82              |



Table 3 details microbenchmark latencies. Audit verification requires only 2.12 ms (p50), while complete 6-phase atomic GST transactions finalize in 30.97 ms (p50) with peak RSS memory under 109 MiB.



**Figure 4: Latency Distribution Across Core GLHS Operations.** Microbenchmark profile showing p50, p95, and p99 percentiles across seven critical transactional and replay operations in the PostgreSQL 16 engine.

### 5.3 Multi-Model Clinical Reasoning Utility

We evaluated model reasoning utility across 1,152 execution cells over a synthetic clinical cohort using state-of-the-art LLMs (Claude 4.6 Sonnet and Gemini 3.6 Flash). Table 4 reports the primary subject-level exact-match contrasts.

**Table 4: Preregistered Multi-Model Utility Contrasts for Strict THSS.** Holm-adjusted significance demonstrates substantial improvements over unconstrained baselines.

| Model             | Comparator         | Effect          | 95% Bootstrap CI   | Holm $p$                  | Discord |
|-------------------|--------------------|-----------------|--------------------|---------------------------|---------|
| Claude 4.6 Sonnet | Naive RAG          | + <b>32.81%</b> | [+21.88%, +43.75%] | $p = 9.54 \times 10^{-6}$ | 21      |
| Claude 4.6 Sonnet | Last-Write-Wins    | + <b>25.00%</b> | [+15.63%, +35.94%] | $p = 2.75 \times 10^{-4}$ | 16      |
| Claude 4.6 Sonnet | Full History       | + <b>18.75%</b> | [+7.81%, +29.69%]  | $p = 1.10 \times 10^{-2}$ | 14      |
| Claude 4.6 Sonnet | Long Context       | + <b>14.06%</b> | [+6.25%, +23.44%]  | $p = 1.95 \times 10^{-2}$ | 9       |
| Claude 4.6 Sonnet | BTSA (Zhao et al.) | +4.69%          | [0.00%, +10.94%]   | n.s.                      | 3       |
| Gemini 3.6 Flash  | Naive RAG          | + <b>25.00%</b> | [+14.06%, +35.94%] | $p = 2.75 \times 10^{-4}$ | 16      |
| Gemini 3.6 Flash  | Last-Write-Wins    | + <b>25.00%</b> | [+15.63%, +37.50%] | $p = 2.75 \times 10^{-4}$ | 16      |
| Gemini 3.6 Flash  | BTSA (Zhao et al.) | 0.00%           | [0.00%, 0.00%]     | exact tie                 | 0       |
| Gemini 3.6 Flash  | Full History       | 0.00%           | [0.00%, 0.00%]     | exact tie                 | 0       |
| Gemini 3.6 Flash  | Long Context       | 0.00%           | [0.00%, 0.00%]     | exact tie                 | 0       |

Strict THSS achieved **98.44%** all-axes exact accuracy with Claude 4.6 Sonnet (63/64 subjects) and **100.0%** with Gemini 3.6 Flash (64/64 subjects), significantly outperforming Naive RAG and LWW baselines ( $p < 0.001$ ).

## 6 Discussion and Systems Implications

### 6.1 Architectural Significance of Disclosure-to-Commit Binding

The empirical results prove that verifying base state versions alone is insufficient in agentic healthcare: without cryptographic binding of the inference disclosure, 100% of adversarial context injection and stale consent replays successfully penetrate the durability boundary. By formalizing THSS and the 6-phase GST kernel, GLHS transforms clinical LLMs from unverified text generators into verifiable, stateful transaction actors.

## 6.2 Trade-Offs in Concurrency Granularity

While fine-grained entity partitioning completely eliminates false-stale contention (0.0%), it requires rigorous lock ordering. The 7-class canonical lock hierarchy proven in our TLA+ specification ensures that even complex multi-entity transactions (such as cross-department clinical council recommendations) execute without deadlocks or cascading aborts.

## 7 Conclusion

We have presented GLHS, a formally verified disclosure-to-commit governance protocol and concurrency kernel for longitudinal health AI. By coupling RFC 8785 canonical Merkle snapshots with a 6-phase GST persistence kernel, a 7-class lock hierarchy, and Wound-Wait DAG preemption, GLHS bridges AI inference and electronic health record persistence. Formal TLA+ model checking across 148 million states and isolated PostgreSQL benchmarks demonstrate 100% invalid commit prevention ( $p = 1.73 \times 10^{-77}$ ), high throughput scaling ( $> 300$  tx/s), and near-perfect clinical reasoning accuracy. GLHS establishes a dependable foundation for the next generation of safe, stateful medical AI agents.

## Statements and Declarations

**Competing interests.** The authors declare no competing financial or non-financial interests.

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**Data and code availability.** Full source code, formal TLA+ proofs, benchmark harnesses, and reproducibility artifacts are open-source in the CLARA-Care repository: <https://github.com/Project-CLARA-HBT/CLARA-Care>.

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